

Unit: 3. Testing.

"Testing" :- It is the process of executing a program with an intent of finding an undiscovered errors. It is used to detect difference b/w given i/p & expected o/p.

Objective of Testing

1. Discover as many error possible in a given time & To check whether the given s/w product matches its requirement specification. ally
2. Validate the quality of s/w using minimum cost & efforts.
3. Generate high quality test cases, perform effective test & helpful in generating problem reports. des n
- 4.

Qualities of a Good Test :-

1. Checks whether system meets requirements & executes successfully in the environment.
2. Good test case is that which has probability of finding errors & yet undiscovered errors.
3. Good test is not redundant.
4. Good test should never be too simple or too complex.

Categorisation of Black & White Box testing :-

White Box testing

- Unit Testing
- Integration Testing
 - Top-Down Approach
 - Bottom up Approach

Black Box testing

- Acceptance Testing
- Regression Testing
 - Alpha
 - Beta

★ Unit Testing :- It's a software development process in which smallest testable part of an application called as Units are individually tested for the proper operation.

★ Integration Testing :- When individual modules are merged & tested as a whole group then it is known as integration testing

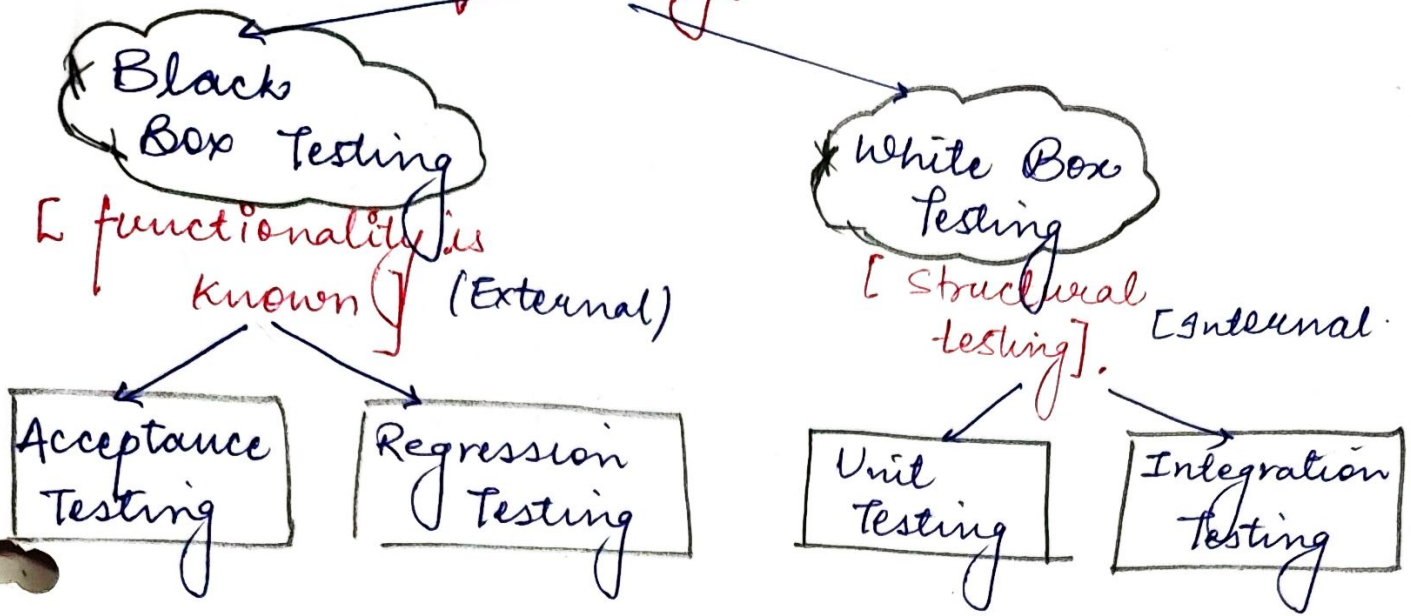
Types of integration testing :-

(A) Top-Down Approach :- When the top integrated modules are tested and the branch of the module is tested step by step until it comes to an end of that related modules.

(B) Bottom up Approach :- When the lowest level components are tested first then it is used to facilitate the higher level component, the process goes on repeating until the module at the top of hierarchy is tested.

Types of Testing

- CFG



Diff B/w Black Box & White Box Testing

Black Box Testing

- ① It is a testing technique in w/c the software is tested without knowing its internal structure code of the program.
- ② Performed by Testers
- ③ Implementation & programming is not required
4. Applied on higher levels like regression & acceptance testing.
5. Aim : what functionality system is performing

White Box Testing

- ① It is testing in w/c internal structure of a software is known.
- ② Performed by developers Required.
- Applied on Unit & integration testing.
- Aim :- How the system is performing through code, structure

Components of Testing :-

- CFG

- Error/Bug/Defect
- Fault
- Failure
- Test Cases
- Verification & Validation

① Error :- When a development team makes a mistake in any phase of SDLC or SWDLC errors are produced

② Fault :- Fault is a condition w/c arises due to error in the program that actual cause a failure.

③ Failure :- It is the inability of the system or component to perform a required function according to its specification.

④ Test Cases :- Test cases are well documented procedure design to test the functionality of the system.

⑤ Verification & Validation :-

Verification :- Are we Building the product right?

Validation :- Have we build the right product?

Alpha & Beta Testing

Alpha testing conducted by team of highly skilled testers at development side

- Dedicated team is required
- Not open for market or public.

→ Done for software application, project and product

→ Conducted within organization

→ Developers & Business Analyst are involved

→ Uses both Black as well as white box.

Conducted in virtual env.

Performed by consumers or end users at their own site.

Done for S/W product like operating system, write a paint utilities outside of organization.

Only Black box.

(%|=length)

(c) Green is used for user reference, statement or comment line.

(d) Brackets appeared in for, end, if...

instead of two colons

(c) Maroon can be seen when putting only one apostrophe (')

(b) Violet is appeared under two apostrophe (")

(a) Black is appeared for coding.

1. COLOURS SIGNIFICANCE IN MARKING

Red line

- CFG

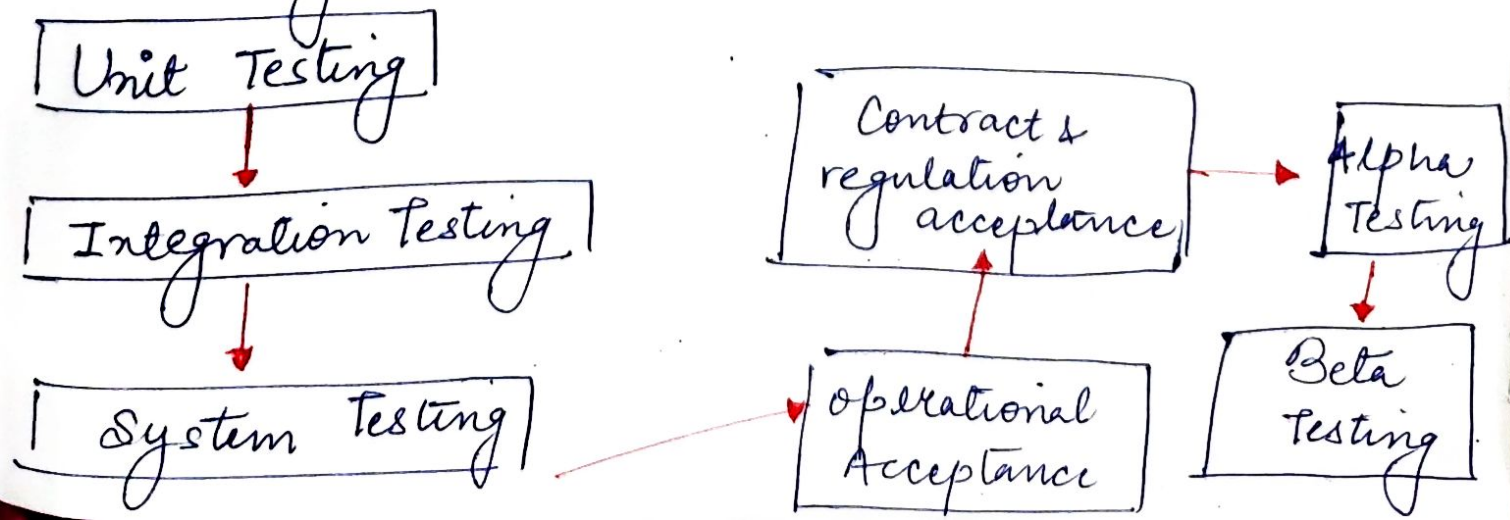
Acceptance Testing :- It is the formal testing to determine whether the software system satisfies its acceptance criteria & enable to determine whether to except or reject the system.

Types :-

- (1) Alpha :- It is test period during w/c the product is complete & usable in the test environment but not necessary it is error free.
- (2) Beta :- Once the alpha phase is complete, the development enters the beta phase, the beta phase is the test period during w/c a product should be complete & usable in production environment.

Beta testing also known as user testing takes place at the end users site by the end users to validate the usability, functionality & compalility testing

Testing in SDLC



Regression Testing :-

- CFG

It is a type of software testing that verifies the software previously developed & tested still performs correctly even after it was changed or interacted with other software.

Purpose :- To ensure that changes such as those mentioned earlier haven't introduced new faults.

Types of Regression Tests :-

Final Regression Tests

is performed to validate the build hasn't changed for a period of time.

Regression Tests

A normal regression testing is performed to verify if the build has NOT broken any ^{other} parts of application by recent code changes.

Cyclomatic Complexity - CFG

It is the Source Code Complexity measurement which is being correlated to a number of coding errors.

It is calculated by developing a Control Flow Graph (CFG) of code that measures the number of linearly independent paths through a program module.

Example -

If A = 10 Then
If B > C Then
A = B

Else
A = C

ENDIF

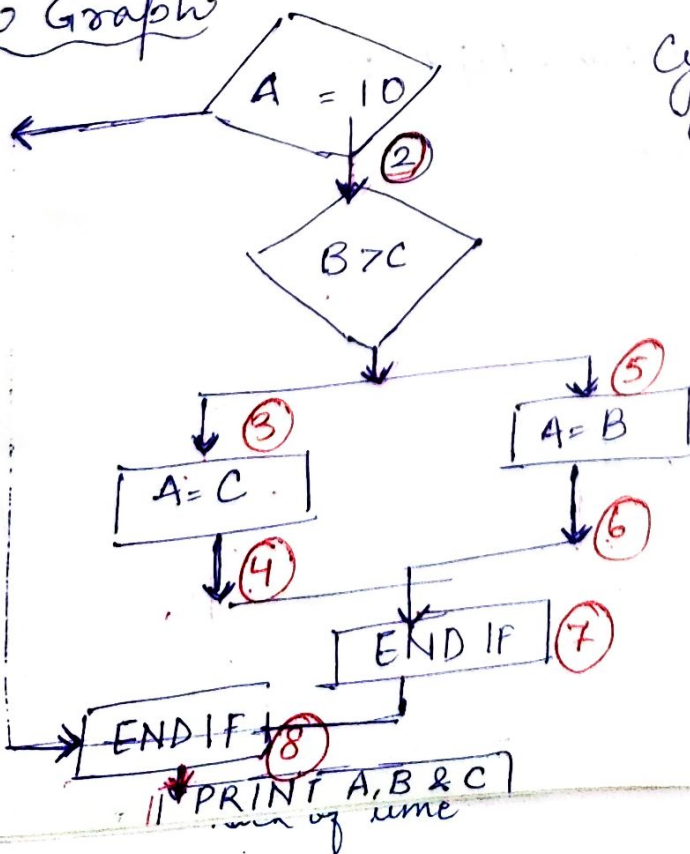
ENDIF

Print A

Print B

Print C.

Flow Graph



Cyclomatic Complexity = $E - N + 2$

E = No. of edges in the flow diagram

N = No. of Nodes.

P = No. of nodes that have exit points

$$N = 7$$

$$E = 8$$

$$V(G) = E - N + 2$$

$$= 8 - 7 + 2$$

$$= 1 + 2$$

$$= \textcircled{3}$$

$$V(G) = P + L$$

$$= 2 + 1 = \textcircled{3}$$

A, B, C which is the maximum among all.

$P =$ Predicate Nodes
denotes total
no. of condition

Here we have 2
condition

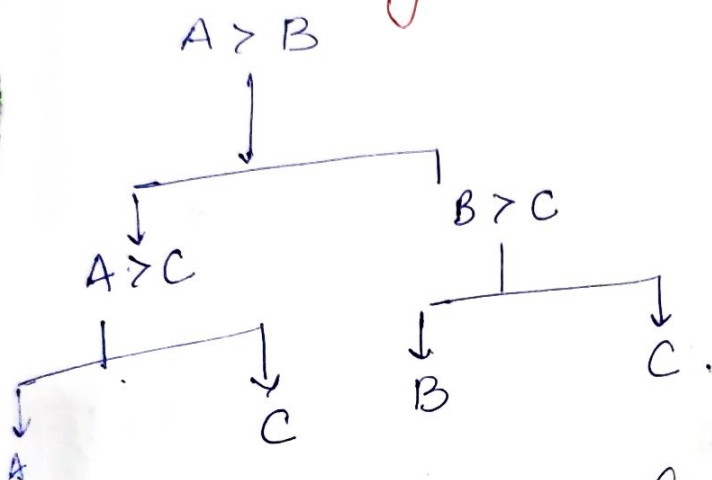
$$A = 10 \text{ \& } B > C$$

$$E = \textcircled{P} + L$$

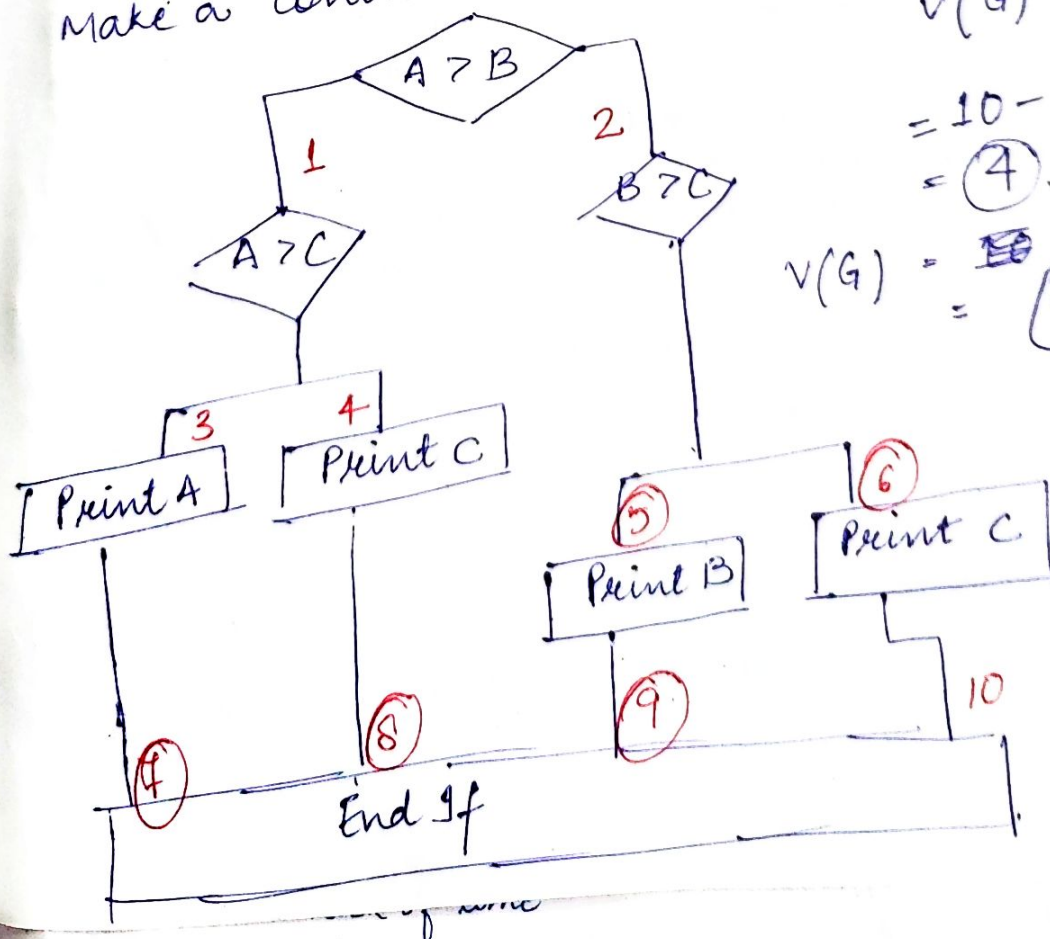
No. .

ist.

lyze



Make a Control flow Graph



$$V(G) = E - N + 2$$

$$= 10 - 8 + 2$$

$$= \textcircled{4}$$

$$V(G) = \textcircled{3} + 1$$

$$= \textcircled{4}$$